

LD4B-850-V-10G-2

OVERVIEW

Laser diode coupled to an optical fiber and packaged into a hermetic case.

MAIN FEATURES

- Wavelength: 850 nm
- Cavity type: VCSEL
- Data rate up to 10 Gbps
- Optical power in CW mode in multimode fiber: 2 mW
- Package types: coaxial, coaxial with bracket
- Built-in monitor photodiode

ORDERING INFORMATION

LD4B-850-V-10G-2-X-X-X-X-X-X

Case type

COAX: compact coaxial (low duty cycle pulse mode only)

COAXB: compact coaxial with a bracket

Other type on request

Pinout code

32: see more details on page 5

Fiber type

MM5: MM, [50/125. OM3](#), furcation tubing Ø0.9 mm

MM6: MM, [62.5/125. OM1](#), furcation tubing Ø0.9 mm

Other type on request

Connector type

FU: FC/UPC (SM05, SMP05)

FA: FC/APC (SM05, SMP05)

N: no connector (scissors cut)

Other type: on request

Test measurements

CW: CW mode (electro-optical parameters at T=25+/-5 C and spectrum)

Fiber length

0.5: 500+/-50 mm

1.0: 1000+/-100 mm

Other length on request

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ABSOLUTE MAXIMUM RATINGS

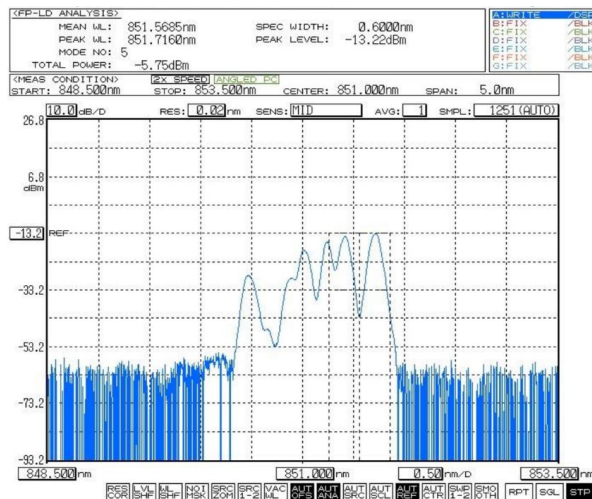
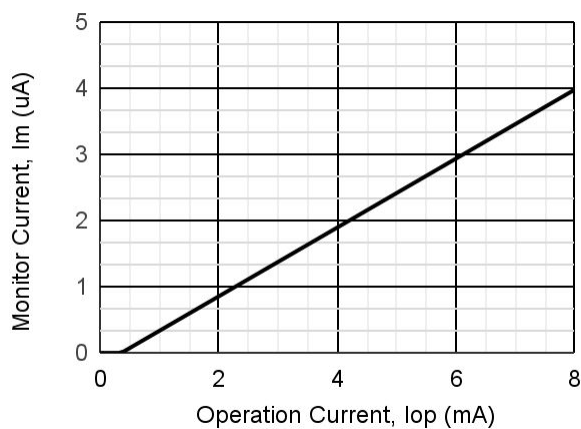
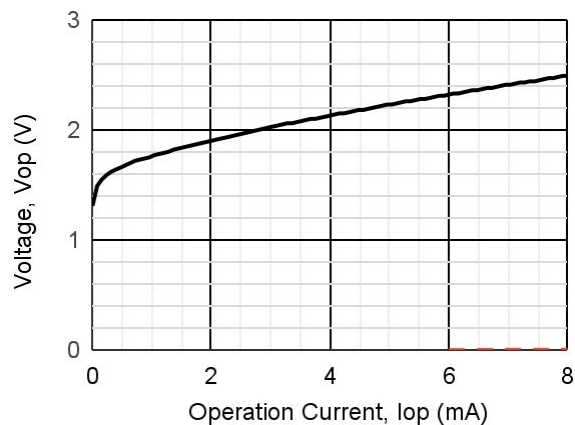
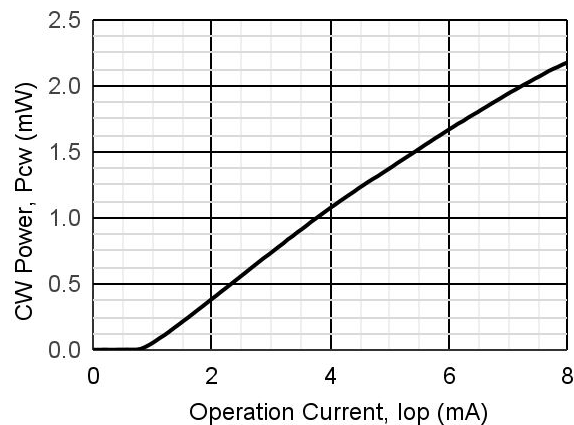
Parameter		Value	Unit	Conditions
Laser diode forward current	$I_{\max}$	10	mA	CW
Laser diode reverse voltage	V_{RL}	10	V	Reverse current 10 μ A
Photodiode reverse voltage	V_{RP}	10	V	
Operating temperature**	T_{OP}	-40 - +85	°C	Coaxial package
Storage temperature	T_{stg}	-40 - +85	°C	
Soldering temperature	T_{sold}	260	°C	Max. 5 seconds

**Operating temperature is defined by the case temperature. It is recommended to ensure sufficient heat dissipation so that the module's maximum operating temperature is not exceeded.

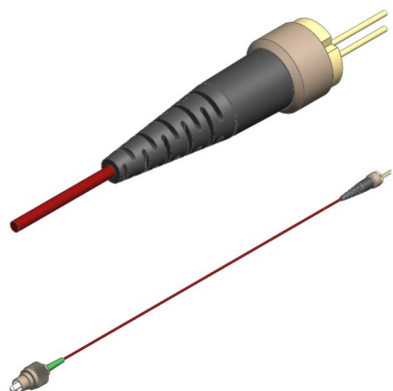
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ELECTRICAL-OPTICAL CHARACTERISTICS (T = 25 °C)

Parameter		MIN	TYP	MAX	Unit	Conditions
Optical power (CW)	P _{cw}	2			mW	CW, I _{op} = 8 mA, MM5
Mean wavelength	λ	840	850	860	nm	CW, I _{op} = 8 mA
Spectral width	Δλ			2	nm	CW, I _{op} = 8 mA
Wavelength-temperature coefficient	dλ/dT		0.07		nm/°C	CW, I _{op} = 8 mA
Threshold current	I _{th}			1.5	mA	
Slope efficiency	S _e	0.20	0.30		mW/mA	CW, MM5
Operating voltage	V _{op}		2.5	2.6	V	CW, I _{op} = 8 mA
Monitor current	I _m	1		7	uA	CW, I _{op} = 8 mA, V _r = 5 V
Total capacitance	C _t		12		pF	V _{rd} = 3 V, f = 1MHz

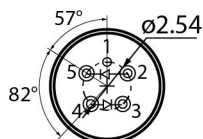
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COAX

BACK VIEW



PINOUT

#32

1. Case
2. LD anode
3. PD cathode
4. PD anode
5. LD cathode

Download more
information



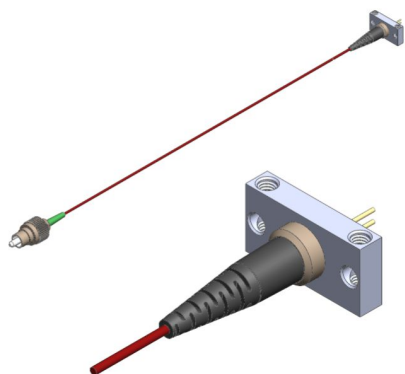
Drawing



3D model

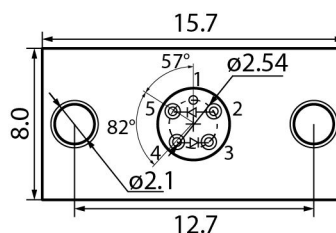


Application Notes



COAXB

BACK VIEW



PINOUT

#32

1. Case
2. LD anode
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Drawing



3D model



Application Notes

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Characteristics, data, materials and structures specified in this datasheet are subject to change without notice. Please refer to the latest specification before use of the products.

Safety and handling cautions

1. Avoid smashing and burning of the module. Avoid storing and using the module in conditions where water, organic solvents or aggressive acids or bases may contact the module or where there is a possibility of exposure to corrosive gases, explosive gases, dust, salinity or other harsh conditions. The module should be disposed as special industrial waste.
2. Exceeding absolute maximum ratings even for a short time can cause permanent damage of the module.
3. The module is sensitive to and can be broken by ESD (static electricity).

Conflict Minerals Policy Statement

LD4B achieves business objectives and customer needs with social responsibility. We do not support or contribute to the violence and human rights violations associated with the mining of conflict minerals coming from Conflict Regions according to US "Dodd-Frank Act". When possible, our suppliers' conflict mineral statements are reviewed. We do not directly purchase Conflict Minerals from any source and do not knowingly procure any parts and products containing Conflict Minerals from Conflict Regions.

RoHS Compliance Statement

Restriction of Hazardous Substances (RoHS) directive (Directive 2011/65/EC amended with Directive (EU) 2015/863) is the directive aimed at reducing the harmful environmental impact of waste electrical equipment by restricting the use of known dangerous substances. Based on information received from our supply sources, LD4B hereby states that the banned substances listed in the RoHS directive are not found in the parts and materials used above the threshold level listed other than exceptions approved by the European Commission.

REACH Compliance Statement

Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) is a European Union regulation 1907/2006/EC that addresses the production and use of chemical substances, and their potential impacts on human health and the environment. Based on information received from our supply sources, LD4B hereby states compliance of the parts and materials used in manufacturing to REACH regulation. LD4B does not manufacture or import any substances or preparations as defined under REACH.